ST. LAWRENCE SECONDARY SCHOOL

CHABAHIL, KATHMANDU

FINAL EXAMINATION – 2078

GRADE: XI	FACULTY: SCIENCE	FM: 75
TIME: 3 HRS	**SUBJECT: CHEMISTRY (3011)**	PM:30

All candidates are requested to answer the following questions in their own language.

Group- 'A'

A. Circle the best alternative to the following questions. **(11x1=11)**

1. What is the weight (in grams) of 0.5 moles of $Ca(OH)_2$?
   a) 47 b) 37
   c) 73 d) 74
2. How much calcium carbonate should be heated to produce 1.12 litres of CO_2 at NTP?
   a) 5 g b) 4 g
   c) 3 g d) 2 g
3. Which of the following has zero dipole moment?
   a) H_2O b) NH_3
   c) SO_2 d) BF_3
4. Which of the following is nucleophilic reagent?
   a) $FeCl_3$ b) BF_3
   c) $AlCl_3$ d) NH_3
5. Formation of prussian blue colouration in the Lassaigne's test indicates the presence of
   a) Sulphur b) Bromine
   c) Nitrogen d) Hydrogen
6. Which of the following is not alkaline earth metal?
   a) Magnesium b) Barium
   c) Beryllium d) Sodium
7. What type of hybridization occurs in the carbon atom of ethene?
   a) SP^3 b) SP^2
   c) SP d) SP^4
8. Efflorescent solids
   a) Absorb moisture
   b) Absorb so much water
   c) Lose its water of crystallization
   d) Become hydrated
9. Which statement is true for one cycle of sodium-potassium pump for every ATP molecule?
   a) 3 Na^+ ions are exported
   b) 2 K^+ ions are imported
   c) Both
   d) None
10. Which of the following is amphoteric oxide?
    a) Na_2O b) CO_2
    c) ZnO d) NO
11. Which gas is obtained by heating white phosphorus with very strong or concentrated sodium hydroxide solution?
    a) HI b) NH_3
    c) PH_3 d) H_2S

Group- 'B'

B. Give short answers to the following questions. **(8x5=40)**

12. An element **X** has 2, 8 and 7 electrons in K, L and M shells respectively.
    i. Idetify X. (1)
    ii. The size of $\mathbf{X}^-$ ion is greater than that of **X** atom though both contain same number of protons, why? (2)

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iii. Mention the group and period in which the element **X** lies in the periodic table. (1)

iv. State Modern Periodic law. (1)

OR

What is meant by oxidation number? Balance the following equation by oxidation number or ion electron method. (1+4=5)

$Zn + HNO_3 \longrightarrow Zn(NO_3)_2 + H_2O + NO$

13. Sodium metal is extracted from its cheapest and the most abundantly available compound sodium chloride by electrolysis process. Difficulties arise during the electrolysis of sodium chloride for a long time.

i. How did J. C. Down overcome these difficulties? (1)

ii. Give the principle and draw well labeled diagram for the extraction of sodium by Down's process. (1.5+1.5=3)

iii. What are the reasons behind that sodium metal cannot be extracted by carbon reduction process? (1)

14. Define law of mass action. Establish the relationship between K_P and K_C, where, K_P and K_C are the equilibrium constant in terms of partial pressure and concentration respectively. Also give any one chemical equation of homogeneous equilibrium where K_P is equal to K_C. (1+3+1)

OR

State Graham's law of diffusion. A vessel of volume 100 ml contains 10% O_2 and 90% unknown gas. The gases diffuse in 86 sec through a small hole of vessel. If pure oxygen under the same condition diffuses in 75 sec, find the molecular mass of the unknown gas. (1+4)

15. Group VIIA elements of the periodic table are called halogens.

i. Name super halogen? (0.5)

ii. Compare bleaching action of SO_2 and Cl_2. (2)

iii. 'HCl has polar character though it has covalent bond', why? (1.5)

iv. Mention the difference between hydrochloric acid and hydrogen chloride. (1)

16. Metallurgy is the process of extraction of metals from their ores. There are different metallurgical processes for extraction of metals.

i. Give the meaning of pyrometallurgy. (1)

ii. What type of ore is concentrated by froth floatation process? (1)

iv. What do you mean by poling? (1)

iii. Mention the differences between calcinations and roasting. (2)

17. One example of homologous series is given below:

$CH_2=CH_2$
X
$CH_3CH_2CH=CH_2$

i. Which class of organic compounds is given in the table? Give its general formula. (1)

ii. Give the structure of **X** with common name and IUPAC name. (2)

iii. What happens when **X** is allowed to react with HBr in presence of organic peroxide? (1)

iv. What is meant by homologous series? (1)

18. Write down the meaning of decarboxylation reaction with one suitable reaction. One organic compound can be converted into another by using organic reactions. Suggest suitable methods for the following conversions? (2+1.5+ 1.5)

i. Carbon to acetylene ii. Benzene to BHC

19. Ammonia is mostly used for the manufacture of urea and this ammonia is manufactured by **Haber's process**.

i. Write down the molecular formula of urea. (1)

ii. Give the principle and draw flow-sheet diagram for the manufacture of ammonia by Haber's process. (1+1.5=2.5)

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iii. What happens when ammonia gas is passed into mercurous nitrate paper? (1)

iv. Mention any one harmful effect of chemical industry. (0.5)

Group- 'C'

Give long answers to the following questions. (3x8=24)

20. 8 gm of pure calcium is treated with 50 gm of pure HCl to give $CaCl_2$ and H_2

i. Which is limiting reactant? Show with calculation. (3)

ii. Calculate the mass of unreacted reactant left over? (1)

iii. What mass of $CaCl_2$ will be formed? (1)

iv. How many moles of water will be produced if the whole H_2 formed in the reaction reacts with O_2? (2)

v. What is the importance of limiting reactant in chemical calculation? (1)

21. Sulphuric acid is manufactured by contact process.

i. Write down four steps of chemical equation for the manufacture of sulphuric acid by contact process. (4)

ii. Why is this process called as contact process? (1)

iii. Give any one chemical reaction in which sulphuric acid acts as dehydrating agent. (1)

iv. Why does H_2SO_4 always act as an oxidizing agent? (1)

v. Draw Lewis structure of H_2SO_4. (1)

OR

Danish Scientist **Neils Bohr** made fundamental contributions to understanding atomic structure based on Plank's quantum theory of radiation.

i. How did Bohr's atomic model explain the origin of hydrogen spectra? (2)

ii. Sketch and name various spectral species observed in the atomic spectrum of hydrogen. (2)

iii. What are the limitations of Bohr's atomic model? (2)

iv. What are the values of n, l and m for 3p-orbitals? (1)

v. State Hund's rule. (1)

22. An alkyl chloride **'X'** undergoes Wurtz reaction to give 2, 3-dimethylbutane as major product.

i. Identify **'X'** with chemical reaction. (2)

ii. What is Wurtz reaction? (1)

iii. What happens when the compound **'X'** is heated with alcoholic KOH? (1)

iv. Give the reactions of the compound formed in (iii) with ozone followed by hydrolysis with Zn/H_2O. (2)

v. You are supplied two similar gas jars. One is filled with ethene while the other with ethyne. Suggest any one suitable chemical test to identify them. (2)

Best of Luck

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